Partial Fractions

Section 1: Distinct Linear Factors

1 Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions $\frac{A}{x-1}+\frac{B}{x+2}$.

a. $A=\frac{4}{3}, B=\frac{11}{3}$

b. $A=4, B=11$

c. $A=1, B=4$

d. $A=2, B=3$

2 Find the partial fraction decomposition of $\frac{1}{x^2-9}$.

a. $\frac{1/6}{x-3}-\frac{1/6}{x+3}$

b. $\frac{1/3}{x-3}+\frac{1/3}{x+3}$

c. $\frac{1}{x-3}-\frac{1}{x+3}$

d. $\frac{1/6}{x-3}+\frac{1/6}{x+3}$

3 Evaluate $\int \frac{3}{(x+1)(x+4)}dx$.

a. $\ln\left|\frac{x+1}{x+4}\right|+C$

b. $3\ln|x+1|+3\ln|x+4|+C$

c. $\frac{1}{3}\ln\left|\frac{x+1}{x+4}\right|+C$

d. $\ln|(x+1)(x+4)|+C$

Section 2: Repeated Linear Factors

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4 What is the correct template for decomposing $\frac{2x+7}{(x-2)^2(x+1)}$?

a. $\frac{A}{x-2}+\frac{B}{(x-2)^2}+\frac{C}{x+1}$

b. $\frac{A}{x-2}+\frac{B}{x+1}$

c. $\frac{Ax+B}{(x-2)^2}+\frac{C}{x+1}$

d. $\frac{A}{(x-2)^2}+\frac{B}{x+1}$

5 In $\frac{3x^2+1}{(x-1)^2}=A+\frac{B}{x-1}+\frac{C}{(x-1)^2}$, find C.

a. 4

b. 3

c. 1

d. 0

6 Decompose $\frac{x}{(x+1)^2}$.

a. $\frac{1}{x+1}-\frac{1}{(x+1)^2}$

b. $\frac{1}{x+1}+\frac{1}{(x+1)^2}$

c. $\frac{x}{x+1}-\frac{1}{(x+1)^2}$

d. $\frac{2}{x+1}-\frac{1}{(x+1)^2}$

Section 3: Irreducible Quadratic Factors

7 What partial fraction form should be set up for $\frac{4x^2+2x+5}{(x-3)(x^2+4)}$?

a. $\frac{A}{x-3}+\frac{Bx+C}{x^2+4}$

b. $\frac{A}{x-3}+\frac{B}{x^2+4}$

c. $\frac{A}{x-3}+\frac{B}{x+2}+\frac{C}{x-2}$

d. $\frac{Ax+B}{x-3}+\frac{Cx+D}{x^2+4}$

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8 In the decomposition $\frac{x^2+1}{(x-2)(x^2+1)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1}$, find A.

a. $\frac{5}{5} = 1$

b. 0

c. 2

d. -1

9 Evaluate $\int \frac{2x+3}{x^2+9} dx$.

a. $\ln(x^2+9) + \arctan\left(\frac{x}{3}\right) + C$

b. $\ln(x^2+9) + 3\arctan(x) + C$

c. $\frac{1}{2}\ln(x^2+9) + \frac{1}{3}\arctan(x/3) + C$

d. $2\ln(x^2+9) + C$

Section 4: Improper Rational Fractions (Long Division First)

10 Before performing partial fractions on $\frac{x^3+2x^2+1}{x^2-1}$, what preliminary algebraic step is mandatory?

a. Polynomial long division to separate into polynomial quotient plus proper remainder

b. Factoring the numerator completely

c. Applying the cover-up method directly

d. Multiplying numerator and denominator by x

11 Perform polynomial long division on $\frac{x^3+2x^2+1}{x^2-1}$. What is the quotient?

a. $x+2$

b. $x-2$

c. x^2+2

d. x

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12 Decompose the remainder term $\frac{x+3}{x^2-1} = \frac{x+3}{(x-1)(x+1)}$.

a. $\frac{2}{x-1} - \frac{1}{x+1}$

b. $\frac{1}{x-1} + \frac{2}{x+1}$

c. $\frac{3}{x-1} - \frac{2}{x+1}$

d. $\frac{1}{x-1} - \frac{1}{x+1}$

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